AKSHITHA KOCHIKA

Data Scientist | Generative AI | Machine Learning | Python | SQL

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PROFESSIONAL SUMMARY

Data Scientist with 3+ years of experience in Data Science, Machine Learning, Predictive Analytics, and Generative AI. Proficient in Python, SQL, statistical analysis, data preprocessing, feature engineering, and predictive model development using Scikit-learn, Random Forest, and XGBoost. Experienced in developing predictive models for employee attrition and audio analytics, along with contributing to GenAI-powered applications involving Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP, and intelligent question-answering systems. Skilled in model optimization, performance evaluation, and developing AI-driven solutions to solve business problems.

WORK EXPERIENCE

Data Scientist

ADP, Hyderabad **Sept 2024 – Present**

- Developed and evaluated employee attrition prediction models using Logistic Regression, Random Forest and XGBoost to identify employees at risk of turnover, supporting retention strategies and contributing to an 18% reduction in attrition and $2.1M in annual rehiring cost savings.
- Contributing to the development of Workmate AI, a Generative AI-powered HR assistant that leverages Retrieval-Augmented Generation (RAG), LLMs and vector databases to deliver context-aware responses to employee queries on HR policies, benefits and leave-related information.
- Extracted and transformed large-scale HR and Payroll datasets from enterprise data sources using SQL queries, joins and API-based data extraction, enabling structured workforce analytics and downstream machine learning workflows.
- Performed data preprocessing, cleansing, validation and feature engineering on workforce datasets, ensuring data consistency, quality and reliability for predictive modeling and business intelligence.

ML Engineer

Hanyaa Technologies, Hyderabad **Oct 2023 – Sep 2024**

- Developed and deployed machine learning models to predict song success (Hit/Flop) using historical audio features, metadata and engagement data, improving predictive performance through feature engineering and model optimization.
- Designed and implemented an audio plagiarism detection system using digital signal processing and machine learning techniques to identify audio similarities and potential copyright infringements.
- Executed end-to-end data processing, including data cleaning, preprocessing, exploratory data analysis (EDA), feature extraction and feature engineering on audio datasets.
- Extracted and engineered audio features, including Mel-Frequency Cepstral Coefficients (MFCCs), spectral features, chroma vectors, tempo and rhythm descriptors to improve model performance.
- Built and evaluated supervised classification models, including Random Forest, XGBoost, Support Vector Machines (SVM) and Logistic Regression, using appropriate performance metrics and model validation techniques.

EDUCATION

Bachelor of Technology (CSE)

MLR Institute of Technology & Management **2019-2023**

CGPA – 8.41/10

SKILLS

- **Programming Languages:** SQL, Python
- **Machine Learning & Deep Learning:** Supervised/Unsupervised Learning, Regression (Linear, Logistic), Tree-based Models (Random Forest, XGBoost, LightGBM), ANN, Clustering (KMeans++, DBSCAN, GMM), Dimensionality Reduction (PCA, t-SNE), Time Series Analysis
- **Generative AI:** RAG Applications, Embeddings, Vector Databases (Chroma DB), Prompt Engineering, LLM Fine-tuning
- **Statistics & Data Analysis:** Descriptive/Inferential Stats, Hypothesis Testing, A/B Testing, Gain/Lift Charts, KS Scores
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-Learn, SciPy, Matplotlib, TensorFlow
- **Tools & Platforms:** Jupyter Notebook, Git, MS Excel, Power BI

PROJECTS

Employee Attrition Prediction | Machine Learning

- Developed predictive models to identify employees at risk of attrition using historical employee data and workforce-related features.
- Performed data preprocessing, exploratory data analysis (EDA), feature engineering, and feature selection.
- Implemented and evaluated Random Forest and XGBoost models using Recall, Precision, F1-score, and ROC-AUC.

GenAI-Powered HR Question Answering System | Generative AI, LLM, RAG

- Contributed to the development of an AI-powered HR question-answering system for retrieving relevant information from HR policies and documents.
- Worked with concepts and components of Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and semantic search.
- Focused on contextual information retrieval and generating relevant responses to employee HR-related queries.

Hit vs. Flop Song Prediction | Machine Learning, Audio Analytics

- Developed a machine learning classification model to predict song popularity using audio signal processing techniques.
- Extracted audio features using Librosa, including MFCC, Chroma, Spectral Centroid, and Zero-Crossing Rate (ZCR).
- Implemented Random Forest classification and GridSearchCV for hyperparameter tuning and model optimization.
- Built an interactive prediction interface using Streamlit.

Song Plagiarism and Similarity Detection | Audio Processing

- Worked on audio similarity analysis to identify potentially copied or similar musical compositions.
- Applied audio feature extraction and signal processing techniques to compare musical characteristics.
- Explored similarity-based approaches for detecting potential audio duplication.